

Table 1: Time breakdown under different batch sizes (lower is better). R3 Official is treated as the baseline, while R3 Free serves as an oracle reference. All experiments are conducted using the Moonlight-16B model on a two-node cluster with 16 GPUs (TP=16).

Metric	Batch Size											
	1000			2000			4000			8000		
	Free	Official	Fast	Free	Official	Fast	Free	Official	Fast	Free	Official	Fast
Step Time	420	442.5	421.7 ($\downarrow 4.7\%$)	789	810.6	801.0 ($\downarrow 1.2\%$)	1519	—	1590	—	—	—
Rollout Time	52.3	62.04	54.8 ($\downarrow 11.7\%$)	105	120.6	103.1 ($\downarrow 14.5\%$)	191.2	199.8	194.0 ($\downarrow 2.9\%$)	—	—	—
Data Process Time	0.25	0.43	19.82	0.57	1.40	35.39	1.23	—	67.94	—	—	—

Table 2: Time comparison under small batch sizes (lower is better). R3 Official is treated as the baseline, while R3 Free serves as an oracle reference. All experiments are conducted using a DeepSeek model with 16 layers on a two-node cluster with 16 GPUs (TP=16).

Metric	Batch Size			
	10		20	
	Free (Oracle)	Official (Baseline)	Free (Oracle)	Official (Baseline)
Step Time	1520	1630 ($\uparrow 7.2\%$)	2655	2719 ($\uparrow 2.4\%$)